

RAJALAXMI CONSTRUCTIONS

WATERPROOFING, STRUCTURAL REHABILITATION, INDUSTRIAL FLOORING AND COATING SYSTEM.

GSTIN: 36BTNPP0589G1Z6

PAN.BTNPP058

To,

Mr. Manjunath

BCIM Engineering Pvt. Ltd

Date: 18.05.2026

Project: LANCO LH10 – LANCO HILLS TECHNOLOGY PARK PVT LTD, Manikonda, Hyderabad – 500089

Work Order No: WOLANLH10004

Subject: Methodology for Waterproofing Works – Toilet, Balcony, Utility, Terrace & OHT Areas

Dear Sir,

With reference to the Work Order WOLANLH10004 dated 18.05.2026, we hereby submit our detailed methodology for the waterproofing works to be carried out at Lanco LH10 Project. All works shall be executed using approved Fosroc products as specified below, strictly as per the manufacturer's specifications and as directed by the Engineer-in-Charge.

1. TOILET, UTILITY & BALCONY – NON-SUNKEN AREA WATERPROOFING

A. Scope of Work

Providing and applying waterproofing for the toilet and utility non-sunken areas by surface method using Fosroc Brushbond RFX elastomeric waterproof coating with screed concrete protection. Only the plan area shall be measured for payment; wall treatment up to 600 mm height is included in the rate.

B. Materials to be Used

Sl.	Material / Product	Purpose
1	Fosroc Brushbond RFX	Two-component elastomeric cementitious waterproof coating
2	Conplast 421 IC (Fosroc)	Waterproofing admixture for screed concrete and plaster
3	Fosroc Conbextra GP2	Cementitious grout for porous area grouting

C. System Build-Up

- 1st Layer – Mother concrete (existing RCC slab)
- 2nd Layer – Fosroc Brushbond RFX @ 2 coats minimum
- 3rd Layer – Screed concrete 12–15 mm thick with Conplast 421 IC admixture

D. Surface Preparation

1. Inspect the entire area and identify all cracks, porous zones, construction joints, and defective concrete.

2. Clean all concrete surfaces free of laitance, dust, dirt, cavities, and projecting nibs using a wire brush/grinder.
3. Remove all flaky materials and weak concrete by wire brushing.
4. Grout all porous areas, honeycombs, and construction joints with Fosroc Conbextra GP2 cementitious grout and allow to cure.
5. Cut a V-groove of 10 mm wide × 10 mm deep along all surface cracks using a grinder/groove cutter. Clean the groove with compressed air, fill with Fosroc Conbextra GP2 mortar flush with the surface, and allow to cure before coating.
6. At all wall-floor junctions, cut a 12 mm × 12 mm groove along the junction line before coving. Clean the groove, fill with Fosroc Conbextra GP2, and allow to cure. Then provide coves/fillets of 50 mm × 50 mm using CM 1:4 mortar admixed with Conplast 421 IC over the treated junction.
7. Pre-wet the surface with clean water before application. Surface should be Saturated Surface Dry (SSD) – no standing water.

E. Application Procedure – Fosroc Brushbond RFX

8. Mix Brushbond RFX powder and liquid components strictly as per the manufacturer's mixing ratio using a slow-speed drill (max 500 rpm) until lump-free consistency is achieved. Mixing ratio: Brushbond RFX Powder 15 kg + Brushbond RFX Liquid 5.088 kg + Water 3 litres (Industrial Pack).
9. Apply the FIRST coat of Brushbond RFX uniformly over the prepared surface using a stiff brush (120–150 mm width), ensuring even coverage across the entire area including wall treatment up to 600 mm height. Allow the first coat to become 'green' (touch dry).
10. While the first coat is still green, apply the SECOND coat of Brushbond RFX in the cross direction to the first coat to achieve uniform waterproof film thickness of minimum 1 mm total. Coverage: approximately 12–14 sqm per Industrial Pack (15 kg + 5.088 kg) at 1 mm thickness, i.e., 0.5 litre/sqm per coat for 2 coats.
11. Extend coating up to a minimum height of 600 mm on all walls to prevent flashing water ingress.
12. After application, allow air curing only. Water curing or curing compounds shall NOT be applied over Brushbond RFX coating.
13. Pre-Ponding: Before applying waterproofing, ensure all drain outlets are sealed/plugged. Conduct a pre-application ponding test for 24 hours to identify any active leaks; repair all identified leaks before proceeding.
14. Post-Ponding: After complete curing of the waterproof coating (minimum 48 hours after final coat), conduct a ponding test by flooding the area with minimum 25 mm depth of water for 24 hours. Check for any seepage or dampness. Clear the water only after the test is satisfactorily completed.

F. Protective Screed Concrete (12–15 mm Thick)

15. After curing of the waterproof coating, lay a screed concrete 12–15 mm thick over the coated surface.
16. Screed concrete mix: CM 1:4 admixed with Conplast 421 IC @ as per manufacturer's dosage.
17. Ensure proper line and level as per instructions of the Engineer-in-Charge.
18. Work includes material shifting, curing, necessary surface cleaning, masking, wire brushing, lead & lift, etc.
19. Cure the screed for a minimum of 7 days.
20. Only the plan area shall be measured for payment.

2. TOILET & UTILITY – SUNKEN AREA WATERPROOFING

A. Scope of Work

Providing and applying waterproofing for sunken areas by surface method. The work comprises two parts: (Part 1) Brushbond RFX coating with Geo Fabric reinforcement and wall plastering; (Part 2) filling of sunken area with screed concrete. Only the plan area shall be measured for payment; wall treatment is included in the rate.

B. Materials to be Used

Sl.	Material / Product	Purpose
1	Fosroc Brushbond RFX	Two-component elastomeric cementitious waterproof coating
2	Conplast 421 IC (Fosroc)	Waterproofing admixture for screed/plaster mix
3	Geo Fabric / Fibre Glass Cloth (30–50 GSM)	Reinforcement mesh embedded between coats
4	Fosroc Conbextra GP2	Cementitious grout for porous areas, pipe outlets, weep holes
5	Non-Shrink Grout (Fosroc)	Bore packing (paid separately)

C. System Build-Up

- 1st Layer – Mother concrete
- 2nd Layer – Brushbond RFX (1st coat) + Geo Fabric embedding
- 3rd Layer – Brushbond RFX (2nd coat over Geo Fabric)
- 4th Layer – CM 1:3 plaster with Conplast 421 IC on walls (600 mm height)
- 5th Layer – Screed concrete (1:2:4) with Conplast 421 IC to fill sunken depth (total 60 mm including 12 mm metal coba)

D. Surface Preparation

21. Clean and prepare the entire sunken area – remove all loose material, dust, laitance, and flaky matter by wire brushing.
22. Identify and treat all porous areas and honeycombs with Fosroc Conbextra GP2 grout.
23. Fix weep holes as required and grout all pipe/drain outlets with Conbextra GP2.
24. Cut a V-groove of 10 mm wide × 10 mm deep along all surface cracks using a grinder/groove cutter. Clean the groove with compressed air, fill with Fosroc Conbextra GP2 mortar flush with the surface, and allow to cure before coating.
25. At all wall-floor junctions, cut a 12 mm × 12 mm groove along the junction line before coving. Clean the groove, fill with Fosroc Conbextra GP2, and allow to cure. Then prepare inverse gola (cove/fillet) of 50 mm × 50 mm using CM 1:4 mortar at all floor-wall junctions over the treated junction.
26. Bore packing of all tie-rod / plumbing penetration holes with non-shrink grout (paid separately as NT item).
27. Pre-wet the surface to Saturated Surface Dry (SSD) condition.

E. Part 1 – Application Procedure: Brushbond RFX with Geo Fabric

28. Mix Brushbond RFX powder and liquid components strictly as per the manufacturer's mixing ratio using a slow-speed drill (max 500 rpm) until lump-free consistency is achieved. Mixing ratio: Brushbond RFX Powder 15 kg + Brushbond RFX Liquid 5.088 kg + Water 3 litres (Industrial Pack).
29. Apply the FIRST coat of Brushbond RFX uniformly over the complete sunken area (floor + 600 mm wall height) using a stiff brush (120–150 mm width).
30. While the first coat is still green/tacky, lay Geo Fabric / Fibre Glass cloth of 30–50 GSM over the wet coating ensuring proper overlaps (minimum 100 mm overlap at joints).
31. Press the Geo Fabric firmly into the wet coating ensuring full embedment without voids or air pockets.
32. Immediately apply the SECOND coat of Brushbond RFX over the Geo Fabric, fully encapsulating the reinforcement mesh. Coverage: approximately 12–14 sqm per Industrial Pack (15 kg + 5.088 kg) at 1 mm thickness for 2 coats.
33. Extend the coating up to 600 mm height on all walls. Apply 12 mm thick CM 1:3 plaster admixed with Conplast 421 IC on the wall portion above the coating.
34. After application, allow air curing only. Water curing or curing compounds shall NOT be applied over Brushbond RFX coating.
35. Pre-Ponding: Before applying waterproofing, ensure all drain outlets are sealed/plugged. Conduct a pre-application ponding test for 24 hours to identify any active leaks; repair all identified leaks before proceeding.
36. Post-Ponding: After complete curing of the waterproof coating (minimum 48 hours after final coat), conduct a ponding test by flooding the sunken area with minimum 25 mm depth of water for 24 hours. Check for any seepage or dampness. Clear the water only after the test is satisfactorily completed and approved by the Engineer-in-Charge.

F. Part 2 – Filling / Screed for Sunken Area

37. After Part 1 curing, fill the sunken area with screed concrete using 12 mm metal aggregate, mix ratio 1:2:4 admixed with Conplast 421 IC of Fosroc as per manufacturer's specification.
38. Total screed thickness: 60 mm (including 12 mm metal coba layer).
39. Include all cutouts, openings, and check for water tightness.
40. Cure the completed screed for a minimum of 7 days.
41. Only the plan area shall be measured for payment.

3. TERRACE WATERPROOFING

A. Scope of Work

Treating all cracks, construction joints and surface defects, followed by applying two coats of Fosroc Brushbond Roofguard Premium solar-reflective elastomeric polyurethane waterproof coating over a primer of Nitoproof WB. The coating shall be extended to the full height of the parapet wall. Only the plan area shall be measured for payment.

B. Materials to be Used

Sl.	Material / Product	Purpose
1	Fosroc Brushbond Roofguard Premium	Single-component Acrylic-based modified PU elastomeric solar reflective waterproof coating
2	Fosroc Nitoproof WB	Primer coat (diluted 1:2 with water)

3	Fosroc Renderoc CS	Polymer modified cementitious repair mortar for cracks / joints
4	Cebex 100 (Fosroc)	Grout admixture for pressure grouting @ 225 gms/bag of cement
5	Fosroc Renderoc Plug	Rapid-setting non-shrink mortar for fixing/sealing PVC nozzles
6	Fosroc Thioflex 600	Polysulphide sealant for expansion joints and crack sealing

C. System Build-Up

- Step 1 – Surface Preparation: Crack/joint treatment with Renderoc CS + pressure grouting with Cebex 100
- Step 2 – Primer: Nitoproof WB diluted 1:2 with water @ 8–10 sqm/litre
- Step 3 – 1st coat Brushbond Roofguard Premium @ 0.5 mm WFT
- Step 4 – 2nd coat Brushbond Roofguard Premium @ 0.5 mm WFT (cross direction)

D. Surface Preparation & Crack / Joint Treatment

42. Inspect the entire terrace area and identify all cracks, construction joints, cold joints, and hollow/patch areas.
43. Open all construction joints by providing a 'U' groove using a groove cutter; remove dust and clean the area thoroughly.
44. For individual surface cracks, cut a V-groove of 10 mm wide × 10 mm deep along the crack using a grinder. Clean with compressed air, fill with Fosroc Renderoc CS mortar flush with the surface, and allow to cure before coating.
45. Fill the prepared 'U' groove and construction joint areas using Fosroc Renderoc CS repair mortar completely as per manufacturer's instruction.
46. For cold joints / construction joints, provide chemical injection pressure grouting: drill 18 mm dia holes, fix PVC nozzles with Renderoc Plug, inject cement slurry mixed with Fosroc Cebex 100 @ 225 gms per bag of cement at required pressure through PVC nozzles.
47. After grouting, cut projected nozzles flush and seal off PVC nozzles with Renderoc Plug non-shrink rapid-setting mortar compound.
48. Fill all surface hollow portions and surface patches with polymer modified cementitious mortar and allow to cure.
49. Clean and degrease the entire surface using wire brush/blower; ensure surface is free of all dust, laitance, oil, and loose matter.

E. Expansion Joint Treatment (Terrace / Edeck / Basement Levels)

50. Check expansion joints (25 mm to 50 mm / 35 mm to 50 mm width) for leakages.
51. Clean the surface and remove the expansion board to 80 mm depth.
52. Refurbish the joint with backer rod and fill with Fosroc Thioflex 600 polysulphide sealant (or Sika approved equivalent).
53. Protect the expansion joint with aluminium sheet 4 mm thick and 200 mm wide.
54. Test for water tightness after completion.

F. Application Procedure – Brushbond Roofguard Premium

55. Stir Brushbond Roofguard Premium using a slow-speed drill for 1–2 minutes to achieve homogenous consistency. Avoid air entrapment. Allow to settle for 1–2 minutes before use. Do not thin the product.
56. After complete curing of crack/joint repairs, apply primer coat of Fosroc Nitoproof WB: mix base and hardener, then dilute with double the quantity of clean water (ratio 1:2 by volume) to obtain a milky white liquid. Apply using roller or brush at a coverage of 8–10 sqm/litre. Allow primer to dry completely.
57. Apply the FIRST coat of Brushbond Roofguard Premium using roller, brush, squeegee, or airless spray at a wet film thickness (WFT) of 0.5 mm per coat at a coverage of 0.5 litre/sqm (20 sqm per 20-litre pail for 2 coats). Ensure uniform coverage.
58. Allow the first coat to dry for 1–5 hours (depending on climatic conditions) before applying the second coat.
59. Apply the SECOND coat of Brushbond Roofguard Premium at 90° to the first coat at the same thickness and coverage, ensuring a minimum total wet film thickness of 1.0 mm is achieved (resulting in 0.5 mm dry film thickness).
60. Extend the coating over the chajja, wall above chajja for a height of 900 mm, and up to 600 mm on either side of the chajja.
61. Extend the coating to the full height of the parapet wall as per manufacturer's specification.
62. Post-Ponding / Flood Test: After curing the membrane for a minimum of 72 hours, conduct a flood test by ponding the terrace to a minimum depth of 50 mm of water for 24 hours. Plug all drains and place barriers to contain the water. Check for any seepage or dampness during and after the test. Proceed to the next stage only after satisfactory test results are approved by the Engineer-in-Charge.
63. Allow complete curing for 7 days before any further works on the surface.

G. Technical Properties – Brushbond Roofguard Premium

Sl.	Material / Product	Purpose
1	Solid Content	50%
2	Tensile Strength	>2 MPa (ASTM D412)
3	Elongation	>250% (ASTM D412)
4	Shore A Hardness	75 (ASTM D2240)
5	Tear Strength	13 N/mm (ASTM D624)
6	Adhesion to Substrate	2 MPa – concrete failure (ASTM D4541)
7	Crack Bridging	2 mm (ASTM C836)
8	UV Resistance	1000 hours – Pass (ASTM D4587)
9	Service Temperature	-10°C to +60°C

4. OHT (OVERHEAD TANK) WATERPROOFING

A. Scope of Work

Providing and applying two coats of Fosroc Brushbond Roofguard Premium solar-reflective polyurethane waterproof coating on the OHT structure after thorough surface preparation. Only the plan area shall be measured for payment.

B. Materials to be Used

Sl.	Material / Product	Purpose
1	Fosroc Brushbond Roofguard Premium	Single-component PU elastomeric waterproof coating
2	Fosroc Nitoproof WB	Primer coat (diluted 1:2 with water)
3	Fosroc Thioflex 600	PU/Polysulphide sealant for U-groove crack treatment

C. Application Procedure

64. Clean the OHT surface thoroughly removing all loose particles, dust, oil, and laitance.
65. Treat all cracks: make a U groove of 5 mm wide × 10 mm deep, apply primer coating over the groove, and fill with Fosroc Thioflex 600 sealant. Allow to cure.
66. For spider cracks, web cracks or shrinkage cracks (more than 1–2 mm), treat from the negative side using two/one-component PU grouting: drill holes, fix mechanical nipples, and inject with high-pressure electric grout pump.
67. Apply primer coat of Nitoproof WB diluted 1:2 with water @ 8–10 sqm/litre. Allow to dry.
68. Apply FIRST coat of Brushbond Roofguard Premium using roller/brush/airless spray at 0.5 mm WFT @ 0.5 litre/sqm. Allow to dry.
69. Apply SECOND coat of Brushbond Roofguard Premium in cross direction at the same rate.
70. Extend coating to the full parapet wall height as per manufacturer's specification.
71. Cure for minimum 7 days.
72. Only the plan area shall be measured for payment.

5. WATERPROOFING FOR EDECK LEVEL RCC WALL @ EXPANSION JOINT AREA

A. Scope of Work

Waterproofing treatment of RCC walls at the expansion joint area at Edeck level using K10 PU Plus two-component polyurethane coating with LANKO 162 primer, including all crack and joint repairs.

B. Materials to be Used

Sl.	Material / Product	Purpose
1	K10 PU Plus	Two-component polyurethane waterproof coating (2 coats)
2	LANKO 162	Primer coat (1 coat)
3	Fosroc Thioflex 600	PU sealant for cracks and construction joints
4	Fosroc Conbextra GP2	Pressure grouting for cracks / cold joints

C. Application Procedure

73. Clean the RCC wall surface and make it free from all loose particles and dust.
74. Make an inverse gola at all corners using CM 1:4 mortar [1 part cement : 4 part M Sand] mixed with Conbextra GP2.
75. Repair all cracks and cold/construction joints using pressure grouting (Conbextra GP2) and seal joints with Fosroc Thioflex 600 PU sealant.
76. Apply one coat of LANKO 162 primer as per manufacturer's specification and allow to dry.
77. Apply TWO coats of K10 PU Plus over the primed surface as per manufacturer's specification.
78. Apply protective cement plaster coat as the final finishing layer.
79. Work includes all necessary materials, labour, tools & tackles, lead & lift etc.
80. The work shall be completed as per the approved methodology and a 10-Year Unconditional Warranty/Guarantee Certificate/Indemnity Bond shall be submitted by the Contractor.

6. SUPPLY, FABRICATION & INSTALLATION OF G.I. SHEET TRAY FOR EXPANSION JOINTS

Supply, fabrication & installation of G.I. sheet tray of 1 mm thick for expansion joint water leakage at Residential Tower Basements – Levels 1, 2 & 3. Average size: 500 mm width × 100 mm depth, wherever necessary as per site requirement. The tray shall be fabricated and fixed with required fixtures to drain the water collected through a pipe of suitable diameter to the nearest rainwater shaft/chamber. Work includes necessary groove cutting to ceiling/walls as per site conditions, all materials, labour, lead & lift etc., as per the instructions of the Engineer-in-Charge.

7. GENERAL REQUIREMENTS & WARRANTY

A. Pre-Work Requirements

- Detailed methodology to be shared and approved by the Project In-Charge before starting any waterproofing work at site.
- All PPE items, power tools, power cables, pumps, etc. are under the scope of M/s. Rajalaxmi Constructions.
- Curing of all treated areas is under M/s. Rajalaxmi Constructions' scope.
- Surface preparation (cleaning from loose materials, grinding/wire brushing, and air blowing) is under M/s. Rajalaxmi Constructions' scope.

B. BCIM Scope of Supply (Free of Cost)

- RMC, M Sand, Cement, Scaffolding, Jhula, Service Lift, Power, Storage Space, and Water.
- Dewatering shall be under BCIM scope.
- Labour shed shall be provided by BCIM free of cost.

C. Measurement

Only the plan area, measured jointly by the Contractor's site engineer and BCIM's site personnel, duly certified by the Project In-Charge / PMC, shall be considered for billing and payment. Wall treatment is included in the quoted rate and shall not be measured separately.

D. Warranty

The Contractor shall submit a 10-Year Unconditional Warranty / Guarantee Certificate / Indemnity Bond on stamp paper from the date of virtual completion of work, as a Tri-Party Agreement between M/s. Rajalaxmi Constructions, M/s. Fosroc Chemicals India Pvt. Ltd., and M/s. BCIM Engineering Pvt. Ltd.

E. Payment

Rs. 2,00,000/- advance along with order; balance against Running Account (RA) Bills as per actual certified measurements. Retention @ 5% shall be held from each RA Bill until defect liability period of 12 months from virtual completion.

Assuring you of our best quality and services at all times.

Yours faithfully,

A. K. Panda
RAJALAXMI CONSTRUCTIONS

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